DO NOT FORGET TO BRING CALCULATORS FOR QUIZ.

Lecture 3.

Suggested Reading : 2.3 - 2.4 (Pai).

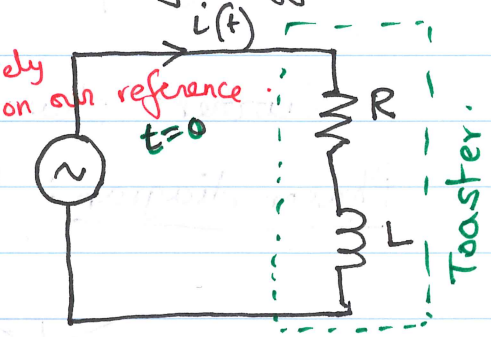
Last time : * Single-phase ac circuits in sinusoidal steady state.
* Concepts of power in ac circuits
  - instantaneous power, $p(t)$
  - Active/Real/Average power, P (W)
  - Reactive power, Q (VAR)
  - Apparent power, S (VA).
* Solved an ac circuit using time-domain approach - solving differential equation.

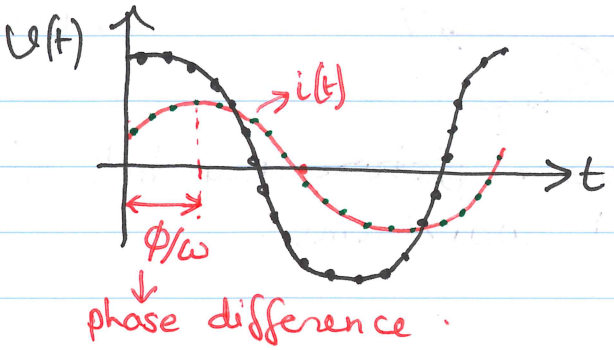
Today : * Phasors and complex power.
* Conservation of power.
* Example 2.5 (Pai).

A. Phasors : Convenient to express and analyse ac circuits in sinusoidal steady state using phasors instead of solving differential equations.

I wrote cos instead of sine. Completely depends on our reference.

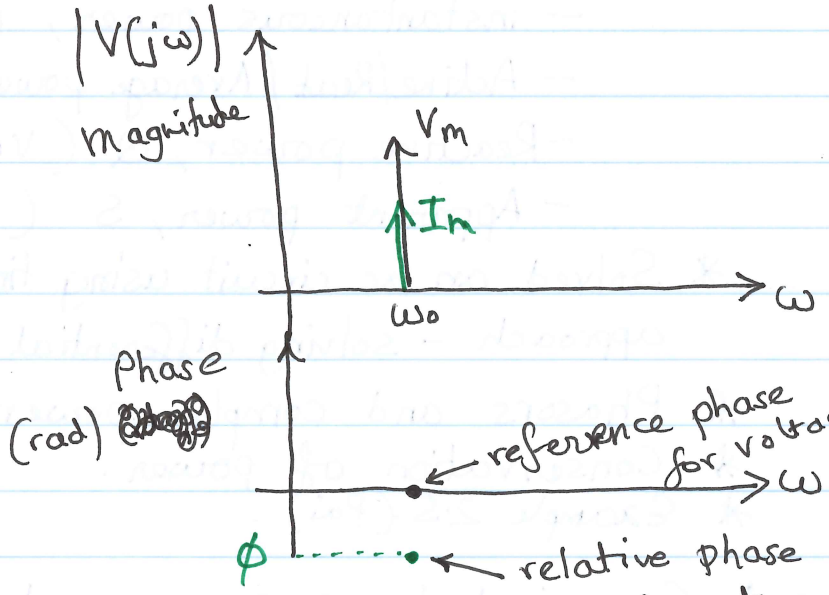
$$v(t) = V_m \cos(\omega_0 t)$$
$$i(t) = I_m \cos(\omega_0 t - \phi)$$





Lot of data points to store information (time domain).

The real relevant information is only magnitude of voltage, current and relative phase. If we do a Fourier transform. and look at frequency spectrum.



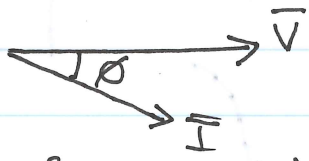
Voltage phasor : $\overline{V} = \left(\frac{V_m}{\sqrt{2}}\right) \angle 0^\circ$ ← reference

→ just a scaling to convert to rms.

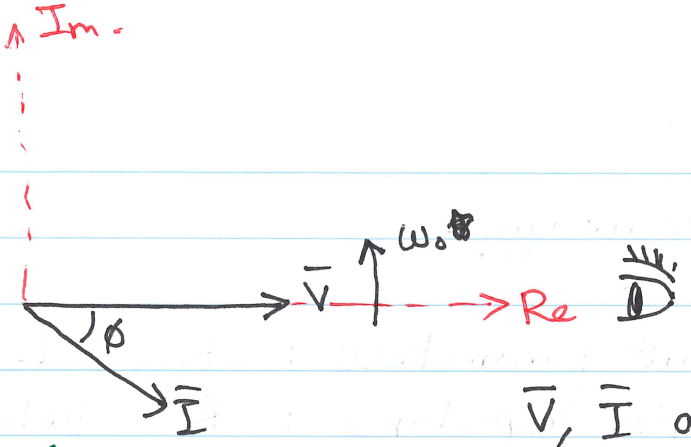
Current phasor : $\overline{I} = \left(\frac{I_m}{\sqrt{2}}\right) \angle -\phi$

↑ negative sign indicates current is lagging the voltage.

Phasor diagram:



A time frozen ~~picture~~ snapshot of the sinusoidal waveforms.



because we chose our reference to be voltage

$\bar{V}$, $\bar{I}$ are complex numbers.

$\bar{V} = V_{rms} e^{j0}$

$\bar{I} = I_{rms} e^{-j\phi}$. } Polar-form.

$= I_{rms}\cos\phi - jI_{rms}\sin\phi$.

So, what is $\bar{V}\bar{I}^*$? $\bar{V}\bar{I}^* = V_{rms}I_{rms}\cos\phi + jV_{rms}I_{rms}\sin\phi$.

$= P + jQ$. (Recall from Lecture 2)

So, complex power,

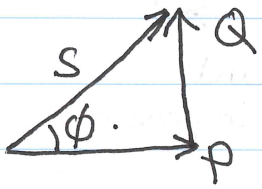
$\bar{S} \triangleq \bar{V}\bar{I}^* = P + jQ$. (unit is VA).

P: active power (a.k.a real, average).

Q: reactive power.

$|\bar{S}| = V_{rms}I_{rms}$ = Apparent power (S).

Power triangle:



Key takeaways :

* Circuit parameters : R L C

  Complex impedance : R $j\omega L$ $\frac{1}{j\omega C}$

  Time domain measurement : $v(t)$ $i(t)$ ϕ ← relative phase.

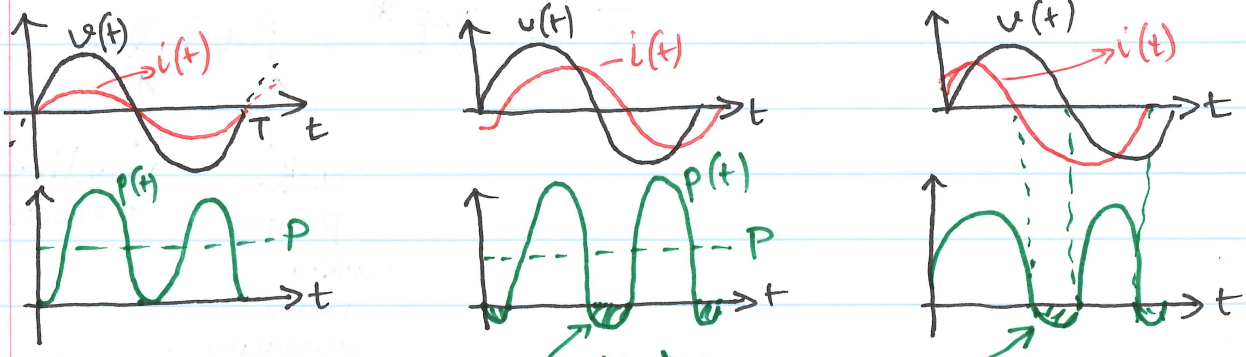
  Phasor domain : $\bar{V}$ $\bar{I}$ $\angle\phi$. power factor → lead / lag.

  Derived quantities : P Q S

  Q (+ve) (-ve).

* R | R-L | R-C

Time



phasor



Voltage and current in phase

Q = 0



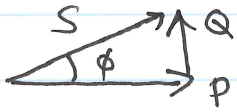
current lagging voltage



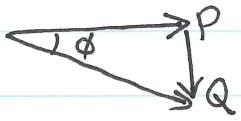
current leading voltage

Power triangle





Q (-)ve



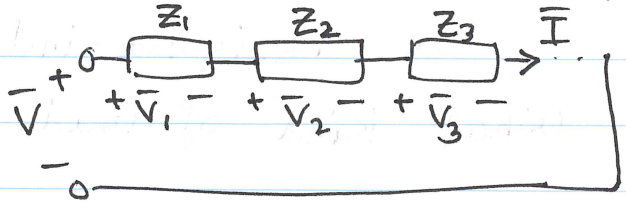
* Conservation of power :

$$\bar{S} = \bar{V}\bar{I}^*$$

A. If multiple loads are connected in series

$\bar{S} = \bar{S}_1 + \bar{S}_2 + \cdots$

(By KVL)

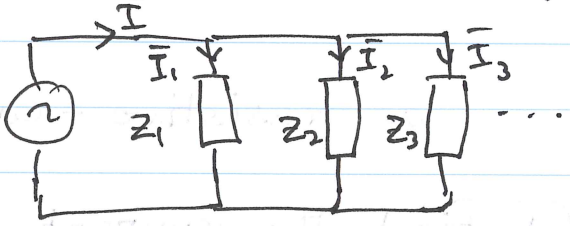


B. If multiple loads are connected in parallel.

$S = \bar{S}_1 + \bar{S}_2 + \cdots$

(By KCL)

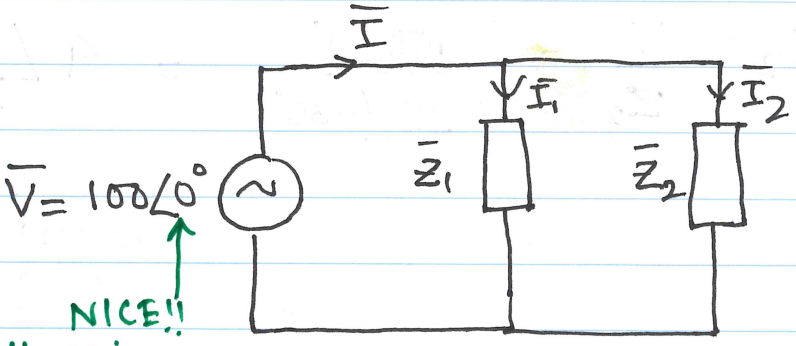
$\bar{I} = \bar{I}_1 + \bar{I}_2 + \cdots$



Bottomline : Complex power is conserved, active power and reactive power are conserved.

Problem 2.5 (Pai)

* First step to attack any problem : Draw the circuit [redacted]. PLEASE !!



$\bar{Z}_1 = 10\angle 45^\circ\ \Omega$

$\bar{Z}_2 = 25\angle -30^\circ\ \Omega$

NICE!! voltage is our reference.

Remember: Complex impedance is a complex number but <u>not a phasor</u>

What kind of load is $\bar{Z}_1$? $10\angle 45°$

polar to rectangular transform of complex no.

$= \underbrace{10\cos 45°}_{+ve} + j\underbrace{10\sin 45°}_{+ve}$

$= R + j\omega L$

a resistive-inductive load.

What kind of load is $\bar{Z}_2$? $25\angle -30°$

$= 25\cos 30° - j\,25\sin 30°$

$= R - \frac{j}{\omega C}$

a resistive-capacitive load.

(a) Find the current in each branch?

$$\bar{I}_1 = \frac{\bar{V}}{\bar{Z}_1} = \frac{100\angle 0°}{10\angle 45°} = 10\angle -45° \text{ A}$$

makes sense current is lagging voltage

UNIT PLEASE! Always

$$\bar{I}_2 = \frac{\bar{V}}{\bar{Z}_2} = \frac{100\angle 0°}{25\angle -30°} = 4\angle 30° \text{ A}$$

makes sense current leads voltage for this branch.

Total current

$$\bar{I} = \bar{I}_1 + \bar{I}_2 = (10\angle -45° + 4\angle 30°)\text{ A}$$
$$= 11.69\angle -25.7° \text{ A}$$

Implies predominantly R-L load overall.

(b) Complex power in each branch?

$$\bar{S}_1 = \bar{V}_0 \bar{I}_1^* = 100\angle 0^\circ \cdot 10\angle +45^\circ \text{ VAR.}$$

Note sign changed because conjugate

$$= \frac{1000}{\sqrt{2}} + j\frac{1000}{\sqrt{2}} \text{ VAR.}$$

P_1 Q_1

makes sense inductive load ≡ positive reactive power.

Similarly,

$$\bar{S}_2 = \bar{V}\bar{I}_2^* = 100\angle 0^\circ \cdot 4\angle -30^\circ \text{ VAR}$$

again sign changed.

$$= 400\frac{\sqrt{3}}{2} - j\frac{400}{2}$$

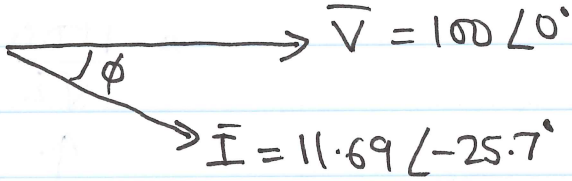
P_2 Q_2

makes sense capacitive load ≡ Negative reactive power.

Total complex power:

$$\bar{S} = \bar{S}_1 + \bar{S}_2 = (1053.51 + j\,507.1) \text{ VAR.}$$

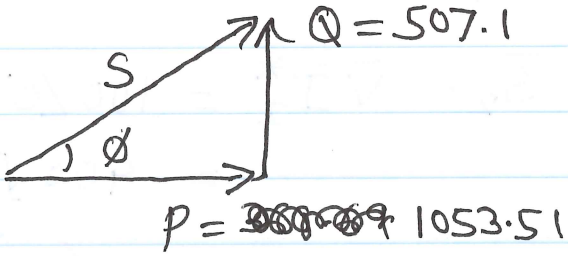
(c) Overall power factor



PLEASE mention this

$$\phi = 25.7° \Rightarrow \cos\phi = 0.9 \text{ (lag)}$$

Let us cross-check with power triangle.



$$\phi = \tan^{-1}\left(\frac{507.1}{1053.51}\right) = 25.7°$$

$$\therefore \cos\phi = 0.9 \text{ (lag)}$$

Q. Impedance of a feeder implies source impedance. (must be connected in series).

